

Tuesday, 3/2/2026

From Venus to Vividh Bharti: More on Damped, Driven Oscillations

1 An Experiment on Venus

Imagine you have a mass attached on a spring which is flown to Venus as part of a mission to better understand the atmosphere of our sister planet. Fun fact: The atmosphere of Venus is an extremely dense, hot, and toxic layer composed of around 96.5% carbon dioxide and around 3.5% nitrogen, with thick sulfuric acid clouds. We wish to get a better understand how dense it is. You, as part of the Venus Mission team, remember you took PYL1002 back in the day and decide to put your damped, harmonic oscillation knowledge to good use.

The experiment consists of a spring of equilibrium length l with a small spring constant k (that is, it's a loose spring). This experimental setup is hung vertically on Venus and the oscillations of a mass m hanging on it are to be studied. The dense atmosphere of Venus has a damping coefficient b , the same one you learnt as $F_{\text{damp}} = -b\dot{x}$. We wish to propose a mechanism to determine the value of b .

- a First, find out the new equilibrium position of the mass hanging from this vertically suspended spring.
- b Now, for small oscillations around this new point of equilibrium, write the equation of motion ($F = ma$) for the mass. You should hopefully have written the equation for a damped harmonic oscillator. Does the natural frequency of the oscillation depend on the strength of gravity g on Venus?

Given the spring is very loose, and Venus has a dense atmosphere, the oscillations are going to be *extremely* overdamped $\gamma \gg 2\omega_0$. You remember from class (yes, from so many years ago now!) that the solution for the overdamped oscillator looks like,

$$x_{\text{overdamped}}(t) = C_1 \exp(-\mu_1 t) + C_2 \exp(-\mu_2 t), \quad (1)$$

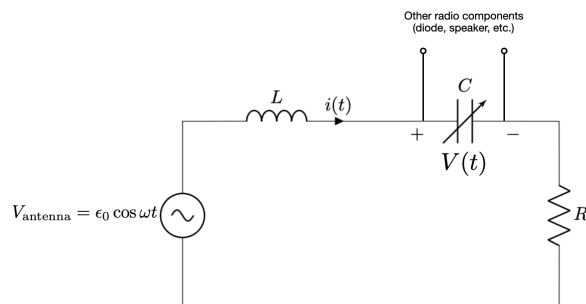
with $\mu_1 > \mu_2 > 0$. You should hopefully remember (or quickly derive) what μ_1 and μ_2 are.

- c In this limit of extremely high damping, show that $\mu_1 \approx \gamma$. Then, find an expression to leading order for μ_2 . You will be tempted to declare μ_2 as zero, and while you're not wrong, you should find the leading, non-trivial dependence on ω_0 and γ .
- d Which term in Eq. (1) (the one with μ_1 or μ_2 ?) is important after a few moments into the motion? Hint: compare μ_1 and μ_2 . *I asked a version of this during the last tutorial as well.*
- e Before moving forward, make a mental picture of how you imagine the oscillation to look like if you begin the mass from rest from a certain non-zero amplitude. Would it be fast oscillations? Sluggish?
- f Now use your approximate solution for $x_{\text{overdamped}}(t)$ by using the approximations from parts (c) and (d), to show that (to leading order), the damping force on the mass is balanced out by the restoring force.
- g And thereby we can devise a way to measure the damping coefficient b of Venus' atmosphere. Argue that if we can measure the mass' position and speed at a given time, we can use this information to infer b .
- h Now visit <https://www.falstad.com/harmonicosc/> and set the following control parameters: (i.) Turn off the driving force, (ii.) make the damping large, and (iii.) the tension small. Initialize the mass from a non-zero amplitude and see its motion. Does it match your expectations and calculations from above?

2 Let's Tune In to the India-Pak Match

In ancient times, before things like MP3s or Spotify or YouTube Music were popular, people got their entertainment by listening to AM radio. Well, there are still some AM stations around, for example, if you tune into AM 1368 in Delhi, you will get All India Radio (AIR)'s Vividh Bharti station. The 1368 means that the frequency of the carrier wave is 1368 kHz.

An AM¹ radio is basically an RLC circuit in series which is “powered” by the oscillating electric field coming from the broadcast station causing the voltage at the antenna to be $V_{\text{antenna}}(t) = \epsilon_0 \cos \omega t$. The capacitor in the circuit is variable, and this is kind of what you tune when you change the station on the AM radio. The point is to be able to “tune” the radio to match the frequency of the station broadcast to be able to pick a strong (think resonance) signal. The wires and other components carry some small resistance R .



- a Use Kirchoff's loop rule in the series RLC circuit to show that the differential equation governing the voltage $V(t)$ on the capacitor follows an equation for a driven, damped oscillator,

$$\ddot{V} + \gamma \dot{V} + \omega_0^2 V = V_0 \cos \omega t. \quad (2)$$

What are the values of ω_0 , γ , V_0 in terms of R , L and C and ϵ_0 .

- b Suppose that you are trying to tune your radio to the India-Pak cricket match, which is being broadcasted at frequency f (you may choose to work with angular frequency $\omega = 2\pi f$ to not deal with pesky factors of 2π) and amplitude ϵ_0 . Find the general expression for the voltage across the capacitor as a function of time assuming a small value of R . What is the amplitude A of oscillation for large t ?
- c For what value of C will we get the largest response? Assuming R is small, what is the value of the C for tuning in?

3 Won't Tolerate Interference

This problem continues from the AM radio above. This is where things become interesting, and I highly encourage the enthusiastic student to work this out.

- a What is the width of the response peak? Since the natural frequency ω_0 is a function of C and vice-versa, you can express your answer in terms of ω_0 instead. That is, how much can you change ω_0 by before the amplitude of response drops to a fraction αA_{max} of the maximum amplitude for a constant α ? It may be helpful to make use of a Taylor series approximation here to get a cleaner answer.
- b Unfortunately, there is a Doordarshan news station that is broadcasting a signal at a frequency $f' = 1.002f$, very close to the cricket station at the same amplitude ϵ_0 . Given the two stations broadcasting, what is the voltage across the capacitor as a function of time now? Since the radio will be left on, we are only interested in the solution for large t . Linearity of the differential equation may help here.
- c To pick up the most signal from the India-Pak match, you set C , and hence ω_0 , to receive the largest (steady state) response A from the cricket station. However, the radio will also pick up noise of amplitude A' from the Doordarshan station. What “quality factor” $Q = \omega_0/\gamma$ is needed in the radio so that noise is at most 1% of the signal? In other words, what does Q have to be so that the $A' \leq 0.01A$? What is the maximum resistance in the circuit you can tolerate to be able to enjoy the cricket match broadcast without interference from Doordarshan? (you can leave your answer in terms of other known, fixed quantities)

¹which stands for *Amplitude Modulated*. Don't worry if it doesn't mean much to you right now. We will discuss more about this in a few weeks.